

Rising Sea levels and its effect on Singapore??

Why are sea levels rising?

1) Thermal expansion:

Oceans absorb over 90% of excess heat caused by global warming, causing the water to expand in volume. Although the expansion of water is small for a given temperature increase, the large surface area of oceans leads to a significant rise in sea levels worldwide.

2) Rising global temperatures:

When land ice like glaciers melt, the water flows into the oceans, increasing the total volume of seawater. The continued loss of land ice is one of the largest long-term threats contributing to future sea level rise.

How has it affected SG?

Due to rising sea levels, Singapore's sea level is estimated to rise up to 1.15m by 2100. When combined with extreme events like high tides, heavy rainfall and storm surges, sea levels could potentially rise to 5m. This means that...

- 30% of Singapore's land and offshore islands will be flooded
- homes, businesses, transport and other aspects of our daily lives can be severely disrupted
- it would contribute to saltwater intrusion, where seawater enters freshwater reservoirs and underground aquifers, threatening Singapore's water supply and ecosystems such as mangroves and coral reefs.
- coastal erosion would worsen, gradually reducing usable land area.



Solutions

Infrastructure refers to engineered structures that protect low-lying coastal areas from rising sea levels and flooding.

- Seawalls
- Breakwaters
- Barrages

Natural ecosystems absorb wave energy, slow water movement, and reduce coastal erosion, helping to protect shorelines from rising sea levels.

- Mangroves
- Coral reefs

Urban adaptation involves planning and designing cities to cope with rising sea levels and reduce the risk of flooding.

- Improved drainage systems
- Elevated buildings

